

(a) directed cycle (b) bidirectional ring

Figure 1: The two networks used in Sections 2.2 and 3. An edge $i \to j$ means that agent i is influenced by agent j. In (a) all edges have weight 1, in (b) all edges have weight 1/2. The dashed self-loop of agent 1, of weight t, is the only modification considered: the other edges of agent 1 are scaled by $1-t$, as marked (equation (3)).

2.2.1 Modelling stubbornness: self-weights and stubborn agents

The self-weight a_{ii} is the weight that agent i gives to its own ~~*current*~~ opinion [4], i.e. to its *current* opinion $x_i(k)$ in (1); it indicates how open the agent is to the opinions of the others [5]. An agent with $a_{ii} = 0$ is open-minded and relies completely on the others; an agent with a positive self-weight changes its opinion more slowly, but it still moves towards the opinions it listens to. In social terms, a self-weight is a form of inertia, not an anchor to a fixed opinion.

Definition 2.4 (Stubbornness [5])**.** An agent is said to be *stubborn* if its opinion remains unchanged independent of the others' opinions.

If the opinions obey the FDG model, then agent i is stubborn, $x_i(k) \equiv x_i(0)$, if and only if $a_{ii} = 1$ [5] (and hence $a_{ij} = 0$ for all $j \neq i$, since A is row-stochastic).

Stubbornness is therefore the extreme case of the self-weight, and there is no intermediate degree of it: an agent with $0 < a_{ii} < 1$ is not stubborn, it only has a positive self-weight.

In the digraph, a stubborn agent has no edges towards the other agents: it is influenced by nobody, while the agents that listen to it keep doing so. Hence it forms a strongly connected component made of a single node, which is a sink of $C(G)$ and is aperiodic because of its self-loop. If, in addition, every agent is influenced by the stubborn agent i, directly or through intermediaries, then i is the only globally reachable node, and Theorem 2.1 gives $w = e_i$: the whole group converges to the initial opinion of the stubborn agent (see also [5]).

To move continuously from no self-weight to stubbornness, we modify only the row of agent 1. Given a row-stochastic matrix A with $a_{11} = 0$, as in both networks of Figure 1, we replace its first row by

$$a_1(t)^\top = (1-t)\, a_1^\top + t\, e_1^\top, \qquad t \in [0,1], \tag{3}$$

where e_1 is the first vector of the canonical basis, leaving the other rows unchanged: agent 1 gives weight t to itself and splits the remaining weight $1-t$ among the agents it listened to before, in the same proportions, as marked in Figure 1. Thus $t = a_{11}$: $t = 0$ is the original network and $t = 1$ makes agent 1 stubborn. In the following we consider $t = 0$, $1/2$ and 1.

2.2.2 Self-weights, stubbornness and convergence

The three values of t correspond to three different cases of the Rosetta stone.

- $t = 0$: both digraphs are strongly connected and periodic, so neither network converges (case (ii)).
- $0 < t < 1$: agent 1 still listens to its neighbours with total weight $1 - t > 0$, so the digraph stays strongly connected, and the self-loop of agent 1 is a cycle of length 1, so the digraph becomes aperiodic. By Corollary 1.14 the matrix is primitive, and the group reaches a consensus (case (i)).
- $t = 1$: agent 1 is stubborn and influences every other agent through the cycle, so it is the only globally reachable node (Section 2.2.1) and the group reaches a consensus on $x_1(0)$ (case (iii)).

3 The Friedkin–Johnsen model

3.1 Introduction to the FJ model

3.1.1 The FJ model

In the FDG model every agent keeps averaging the opinions it listens to, and a persistent disagreement can only come from the structure of the network, i.e. from several closed groups (Theorem 2.3). The Friedkin–Johnsen (FJ) model adds a second ingredient: each agent can also be tied to a fixed opinion of its own, called its *prejudice*. Following [5], the model reads

$$x(k+1) = \Lambda W x(k) + (I - \Lambda)\, u, \tag{4}$$

where

- $W \in \mathbb{R}^{n\times n}$ is a row-stochastic matrix, the *matrix of social influences*, i.e. the weighted adjacency matrix of the influence digraph G (Definition 1.1), which plays the role of A in the FDG model;
- $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$ the *susceptibility* of agent i to social influence;
- $u \in \mathbb{R}^n$ is a constant vector collecting the agents' *prejudices*.

Componentwise,

$$x_i(k+1) = \lambda_i \sum_{j=1}^{n} w_{ij}\, x_j(k) + (1 - \lambda_i)\, u_i. \tag{5}$$

At every step agent i averages the opinions it listens to, exactly as in the FDG model, and mixes this average with its prejudice: the weight λ_i goes to the others and the complement $1-\lambda_i$ goes to u_i. The two quantities play different roles: the prejudice u_i is an opinion and says *where* the agent is anchored, while $1 - \lambda_i$ is a weight and says *how strongly*~~. The model with $u = x(0)$ appears in [4], where λ_i is called the openness level~~; we call $1 - \lambda_i$ the *attachment* of agent i ~~and $1 - \lambda_i$ its attachment to its initial opinion, which plays the role of the prejudice.~~ to its prejudice.

The special cases of (4) are the following [5].

- If $\lambda_i = 1$, agent i follows the FDG rule. If $\Lambda = I$, the FJ model reduces to the FDG model with $A = W$; for this reason we assume $\Lambda \neq I$ from now on.
- If $\lambda_i < 1$, agent i is called *prejudiced*: it is anchored to its prejudice u_i.
- If $\lambda_i = 0$, then $x_i(k) = u_i$ for all $k \geq 1$: the agent ignores the others and jumps to its prejudice after one step. Its opinion never changes, i.e. the agent is stubborn in the sense of Definition 2.4, only if in addition $u_i = x_i(0)$. A stubborn agent of the FJ model is therefore an agent with $\lambda_i = 0$ and $u_i = x_i(0)$.

In social terms, the self-weight of the FDG model and the prejudice of the FJ model are two different forms of resistance to influence. A self-weight ties an agent to its *current* opinion, which still moves towards the others (Section 2.2.1); a prejudice ties it to a *fixed* opinion, which never moves.

In the original formulation of Friedkin and Johnsen the prejudices coincide with the initial opinions, $u = x(0)$, and the susceptibilities are coupled to the self-weights through $1 - \lambda_i = w_{ii}$ ~~[5, 4]. As~~ [5]. As in [5], we do not impose these conditions: u and $x(0)$ are independent, and so are Λ and W. The choice $u = x(0)$ will be needed in Section 3.1.4 to define the influence centrality.

3.1.2 Convergence and stability

The FJ model (4) is a linear system with the constant input u. Following [5], we say that it *converges* if the limit $x(\infty) = \lim_{k\to+\infty} x(k)$ exists for every initial condition $x(0)$, and that it is *asymptotically stable* if ΛW is convergent, i.e. $\rho(\Lambda W) < 1$ (Definition 1.8 and Theorem 1.9). Regular matrices are recalled in Definition 1.10.

Both properties are characterized by distinguishing the agents that are reached by the prejudices from those that are not.

With $r = n$, (8) gives, for every $x(0)$,

$$x(\infty) = Vu, \tag{9}$$

with V as in Corollary 3.3.

The condition of Corollary 3.3 can also be read on the matrix ΛW with the tools of the course. The matrix ΛW is no longer row-stochastic: since W is row-stochastic, the i-th row of ΛW sums to λ_i. Rows of agents that are not prejudiced sum to 1, rows of prejudiced agents sum to less than 1, and since $\Lambda \neq I$ at least one such row exists. Hence ΛW is row-substochastic (Definition 1.15) and Lemma 1.16 applies.

For the matrix ΛW, the weighted out-degree of agent i is λ_i: the nodes with out-degree 1 are the agents that are not prejudiced, and the nodes with out-degree less than 1 are the prejudiced agents. The condition of Lemma 1.16 therefore reads: every agent that is not prejudiced is influenced, directly or through intermediaries, by a prejudiced agent, i.e. all the agents are P-dependent. This is exactly the condition of Corollary 3.3. Indeed, [5] remarks that Corollary 3.3 can be turned into a criterion of Schur stability for substochastic matrices~~, and [4] proves the same condition for the FJ model through this lemma.~~.

Two observations give an intuition for Corollary 3.3. First, letting $k \to +\infty$ in (4) shows that the limit $x(\infty) = Vu$ is an equilibrium, $x(\infty) = \Lambda W x(\infty) + (I - \Lambda)u$. Letting $e(k) = x(k) - x(\infty)$ and subtracting this equation from (4) gives

$$e(k+1) = \Lambda W\, e(k), \tag{10}$$

an autonomous linear system in which the prejudice u does not appear: ΛW decides whether the opinions converge, u decides where.

Second, the linear system (10) shows the role of the P-dependent agents. At each step, agent i keeps a fraction λ_i of the weighted average of the deviations of the agents it listens to. An agent that is not prejudiced ($\lambda_i = 1$) only averages the deviations, while a prejudiced agent loses a fraction $1 - \lambda_i$ of them (in (4), this fraction is replaced by the agent's fixed prejudice). The deviations therefore leave the group only through the prejudiced agents, and they vanish only if every agent is influenced, directly or through intermediaries, by one of them, i.e. all agents are P-dependent. A group of agents that is not influenced, even indirectly, by any prejudiced agent loses no deviation and behaves as an FDG model, as in (7).

In social terms, the group converges, for every initial opinion, as soon as every agent that is not anchored to a prejudice listens, directly or through intermediaries, to someone who is. On a strongly connected network, such as the two networks of Figure 1, a single prejudiced agent is enough. Note that this does not require aperiodicity, unlike the FDG model of Section 2.2.

If some agents are P-independent, the model is not asymptotically stable, but it may still converge. Theorem 3.2 gives the condition in terms of W^{22}; in terms of the whole matrix ΛW it reads as follows.

Corollary 3.4 ([5]). *The model* (4) *converges if and only if* ΛW *is regular, i.e. the limit* $\lim_{k\to\infty}(\Lambda W)^k$ *exists.*

As remarked in [5], this property is not trivial: a system $x(k+1) = Ax(k) + Bu$ with a regular matrix A may not converge, and may even have unbounded solutions, as for $A = B = I$.

The decomposition (6)–(7) also explains these results on the spectrum. Listing the P-dependent agents first, ΛW is block triangular,

$$\Lambda W = \begin{bmatrix} \Lambda^{11} W^{11} & \Lambda^{11} W^{12} \\ 0 & W^{22} \end{bmatrix}, \qquad \operatorname{spec}(\Lambda W) = \operatorname{spec}(\Lambda^{11} W^{11}) \cup \operatorname{spec}(W^{22}),$$

and $\rho(\Lambda^{11} W^{11}) < 1$ by Theorem 3.2. Hence the eigenvalues on the unit circle, which decide convergence and asymptotic stability (Theorem 1.9), can only come from W^{22}, i.e. from the FDG model of the P-independent agents: the model is asymptotically stable when there are no P-independent agents ($r = n$), and it converges when the eigenvalue 1 of W^{22} is semi-simple and its other eigenvalues lie inside the unit disk, i.e. when W^{22} is regular.

3.1.3 How the opinions converge: the matrix V and the augmented graph

When all agents are P-dependent, the final opinions are given by (9), and the convex combination of Theorem 3.2 involves only the prejudices:

$$x_i(\infty) = \sum_{j=1}^{n} v_{ij}\, u_j.$$

Three facts follow.

- The initial opinions are forgotten: $x(0)$ does not appear in $x(\infty)$.
- Only the prejudices of prejudiced agents count: if $\lambda_j = 1$, the j-th column of $I - \Lambda$ is zero, hence so is the j-th column of V, and u_j has no weight in any final opinion.
- Every final opinion lies between the smallest and the largest prejudice of the prejudiced agents.

The comparison with the FDG model is instructive. There, the consensus value $w^\top x(0)$ is a weighted average of the initial opinions, with the same weights w for every agent (Theorem 2.1); here, the final opinions are weighted averages of the prejudices, and each agent has its own weights, given by its row of V.

In the FDG model, consensus is a property of the model: the model reaches a consensus if the final opinions coincide for any initial condition $x(0)$ [5]. In the FJ model with all agents P-dependent, the limit $x(\infty) = Vu$ does not depend on $x(0)$ (equation (9)): whether the final opinions coincide is decided by the prejudices u, which are an external input, through the matrix V, and no longer by the initial opinions. ~~In our reading,this is why~~Accordingly, [5] studies the FJ model in terms of convergence and of the form of the limit, and not in terms of consensus.

The notion of consensus of the FDG model applies again only to the P-independent agents ($0 < r < n$), which evolve as an FDG model with matrix W^{22} and whose limit depends on their initial opinions $x^2(0)$ (Section 3.1.2). The P-dependent agents converge to convex combinations of the prejudices and of the final opinions of the P-independent agents (Theorem 3.2), so they do not necessarily end up at the consensus value of the P-independent agents, even when there is one.

The limit (9) has a direct interpretation in terms of the FDG model. Following [5], the FJ model (4) may be considered as an augmented FDG model with n virtual stubborn agents, anchored at the prejudices: the virtual agent i' associated with agent i keeps the opinion $x_{i'}(k) = u_i$ for all k. Explicitly, stacking the opinions x of the real agents and x' of the virtual agents,

$$\begin{bmatrix} x(k+1) \\ x'(k+1) \end{bmatrix} = \begin{bmatrix} \Lambda W & I - \Lambda \\ 0 & I \end{bmatrix} \begin{bmatrix} x(k) \\ x'(k) \end{bmatrix}, \qquad x'(0) = u. \tag{11}$$

Row i of the matrix in (11) sums to $\lambda_i + (1 - \lambda_i) = 1$, and row i' sums to 1; hence the matrix is row-stochastic and (11) is an FDG model on $2n$ agents. In its digraph, which we call the *augmented graph*:

- a real agent i is influenced by a real agent j with weight $\lambda_i w_{ij}$;
- a real agent i is influenced by its virtual agent i' with weight $1 - \lambda_i$, so this edge exists only if agent i is prejudiced;
- a virtual agent i' is influenced only by itself, with weight 1, so its opinion never changes: it is stubborn (Definition 2.4).

Figure 4 shows the FJ model on the directed cycle of Figure 1(a) when agent 1 is the only prejudiced agent, and the corresponding augmented graph. In panel (a) the prejudice is drawn as an input entering agent 1, since it is not an agent of the network but an external piece of information; in panel (b) it becomes the virtual agent $1'$, and the edge follows our convention: it goes from 1 to $1'$, because agent 1 listens to $1'$. In social terms, a prejudice behaves as an additional agent that never changes its opinion: a prejudiced agent keeps listening to the others, and also listens to this stubborn agent placed beside it.

input, with a different equilibrium Vu. In the augmented model u is part of the state: changing it is a perturbation of the initial condition along the directions of the eigenvalue 1, and it is never forgotten; perturbing only the real agents, with $x'(0)$ unchanged, the augmented model forgets the perturbation exactly as the FJ model. Section 3.2.1 reads the perturbation experiment of Figure 7 in these terms.

3.1.4 Influence centrality

In the FDG model the social power w is defined only when the group reaches a consensus (Section 2.1.2), which is not the relevant notion for the FJ model (Section 3.1.3). Nor does the augmented FDG model (11) help: its graph has at least n sinks, so for $n \geq 2$ it never reaches a consensus and its social power is not defined. A different measure is needed, and [5] discusses an extension of the social power to the FJ model, introduced by Friedkin.

Influence centrality. As in [5], we confine ourselves to the asymptotically stable FJ model with $u = x(0)$, the traditional assumption of the Friedkin–Johnsen theory (Section 3.1.1); then $x(\infty) = Vx(0)$ by (9). The assumption $u = x(0)$ is what gives the initial opinions a role: in general $x(\infty) = Vu$ does not depend on $x(0)$, so the initial opinions would have no weight in the final opinions. The social power of agent i is generalized as the mean weight of its initial opinion in determining the group members' final opinions [5]. These mean weights are the elements of the non-negative vector c such that

$$c = n^{-1}V^\top \mathbf{1}_n, \qquad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i(\infty) = \frac{1}{n}\mathbf{1}_n^\top V x(0) = c^\top x(0), \tag{12}$$

and c_i is called the *influence centrality* of agent i [5]. In terms of V, c_i is the average of the i-th column of V, whose entries are the weights of $x_i(0)$ in the final opinions of the single agents. Since V is stochastic, c is non-negative and $c^\top \mathbf{1}_n = 1$, like w (Theorem 2.1)~~, so the two measures can be compared directly~~: both distribute a total weight 1 among the agents. Similar to French's social power, the influence centrality is generated by an opinion formation mechanism [5].

The relation with the social power is made precise in [5]. The convergence and the consensus of the FDG model are equivalent, respectively, to the regularity and the full regularity of the stochastic matrix A [5] (Definition 1.10). For a fixed W, let $\Lambda = \alpha I_n$ with $\alpha \in (0, 1)$, i.e. the same susceptibility for all agents, and consider the corresponding matrix V_α and vector c_α. For $\alpha = 0$, i.e. $\Lambda = 0$, every agent keeps its initial opinion, $V = I$ and $c_0 = n^{-1}\mathbf{1}_n$: the social power is uniformly distributed between all agents [5].

Lemma 3.5 ([5]). *Let W be a fully regular matrix and w stand for the vector of French's social power, corresponding to the FDG model* (1) *with $A = W$. Then, $w = \lim_{\alpha \to 1-0} c_\alpha$.*

The limit cannot be replaced by $\alpha = 1$: for $\Lambda = I$ the FJ model is the FDG model, which is not asymptotically stable, and $I - W$ is singular since $1 \in \mathrm{spec}(W)$ (Theorem 1.7), so V is not defined. In social terms, the influence centrality moves from the uniform distribution, when every agent is anchored to its own initial opinion, to French's social power, when the weight of the prejudices vanishes.

3.2 Stubbornness and influence centrality in the FJ model

Unless stated otherwise, the examples of this section use as matrix W the two networks of Figure 1 without self-loops, the initial opinions $x(0) = (0,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 1)^\top$ of Section 2.2, and prejudices equal to the initial opinions, $u = x(0)$, as is traditional in the Friedkin–Johnsen theory (see [5]). This choice compares the two models on the same data, and it is required by the influence centrality of Section 3.1.4; each figure states its own exceptions.

3.2.1 Modelling stubbornness: prejudices and stubborn agents

Stubborn agents in the two models. A stubborn agent of the FJ model is an agent with $\lambda_i = 0$ and $u_i = x_i(0)$ (Section 3.1.1). As in the FDG model, stubbornness is the extreme case of a graded mechanism, here the prejudice: an agent with $0 < \lambda_i < 1$ is prejudiced but not stubborn, since its

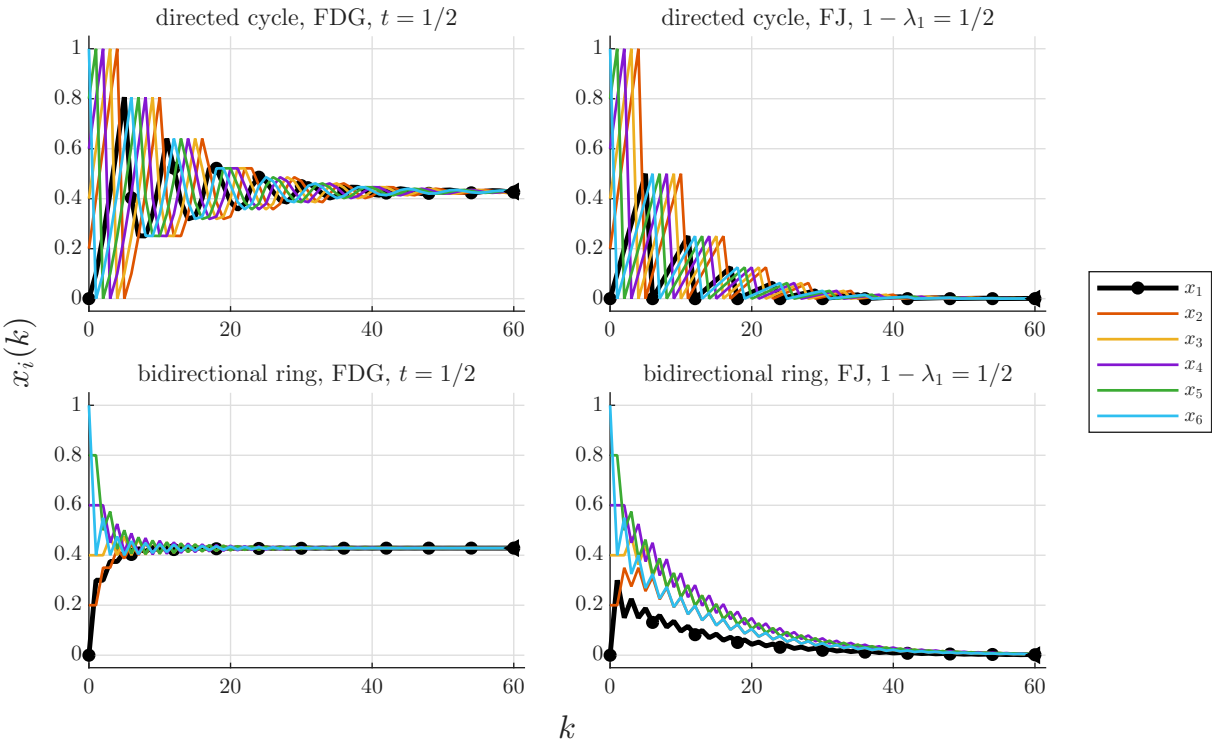


Figure 9: Opinions $x_i(k)$ with the same weight $1/2$ on agent 1, with $x(0) = (0,\, 0.2,\, 0.4,\, 0.6,\, 0.8,\, 1)^\top$ and $K = 60$. Left: FDG model with self-weight $t = 1/2$ (matrix $A(1/2)$, current opinion); right: FJ model with $\lambda_1 = 1/2$, $\lambda_i = 1$ for $i \neq 1$ and $u = x(0)$ (attachment $1 - \lambda_1 = 1/2$, initial opinion). Top: directed cycle; bottom: bidirectional ring. The triangles mark the limits: the consensus value ≈ 0.4286 of the FDG model and $u_1 = 0$ for the FJ model.

Both networks are strongly connected, so every agent is connected by a walk to agent 1: all the agents are P-dependent and, by Corollary 3.3 (case $\Lambda \neq I_n$, W irreducible), the model is asymptotically stable, although the same networks oscillate forever in the FDG model without self-loops. Since column j of V is zero when $\lambda_j = 1$ (Section 3.1.3) and V is stochastic (Theorem 3.2), $V = \mathbf{1}_6 e_1^\top$ and $x(\infty) = u_1 \mathbf{1}_6 = 0$ for every attachment $1 - \lambda_1 > 0$: the group reaches a consensus on the prejudice of agent 1, and crosses the FDG compromise 0.4286 on its way. The equilibrium shows why: agents $2, \ldots, 6$ copy, or average, the agents they listen to, so at an equilibrium they all take the value of agent 1, whose equation becomes $x_1^* = \lambda_1 x_1^* + (1 - \lambda_1) u_1$, i.e. $(1 - \lambda_1)(x_1^* - u_1) = 0$. The fixed value u_1 makes the equilibrium unique, and the attachment only changes how fast it is reached; in the FDG model, instead, every consensus is an equilibrium, and the initial opinions decide which one is reached. In the augmented graph (Section 3.1.3) the ~~real agents listen only to the stubborn virtual agent~~ only virtual agent that the real agents reach, directly or through intermediaries, is $1'$, ~~as in~~ which plays the role of the stubborn agent of Section 2.2.1~~—~~: the whole group converges to its value u_1. In social terms, a self-weight is inertia, which slows an agent down but lets it follow the group, while a prejudice is an anchor: a single anchored agent, however loosely, decides where the whole group ends.

Every agent prejudiced. In Figure 10 every agent resists the influence of the others with the same weight 0.2, in both models: in the FDG model (left) as a self-weight on its current opinion, $A = 0.2\, I + 0.8\, W$; in the FJ model (right) as an attachment to its initial opinion, $\Lambda = 0.8\, I$ and $u = x(0)$. The triangles mark the limits.

On the directed cycle,

$$A = 0.2\, I + 0.8\, W = \begin{bmatrix} 0.2 & 0.8 & 0 & 0 & 0 & 0 \\ 0 & 0.2 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.8 & 0 \\ 0 & 0 & 0 & 0 & 0.2 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0 & 0.2 \end{bmatrix}, \qquad \Lambda W = 0.8\, W = \begin{bmatrix} 0 & 0.8 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0 & 0 \end{bmatrix};$$